Sandra Yang

aleyang@stanford.edu | sandrayang.me | linkedin.com/in/aleyang | github.com/birdhumming | @sandylikesfrogs

Education

Stanford University Jun. 2028
B.S. Computer Science + Mathematics *Stanford, CA*
- Coursework: Machine Learning, Reinforcement Learning, Operating Systems, Algorithms, Probability, Statistical Inference, Stochastic Processes, Real Analysis, Combinatorics
- Co-Director & Lead Problem-Writer of Stanford ProCo; Counterpoint A Cappella; Women in National Security

Experience

Ramp Jun. 2026 – Sep. 2026
Software Engineer Intern *San Francisco, CA*
- Early engineer on Ramp Router, an LLM gateway that cuts inference costs ~40%: helped design and build the initial gateway, onboarding, and attribution, and wrote the first intelligent routing strategies (semantic difficulty-based model selection).
- Co-ran BugBasher, a four-week experiment giving an autonomous agent (Devin) a Ramp card, phone, and email to run a B2B business from a cold start; built the heartbeat automations, shared "brain" memory store, and voice pipeline (Twilio, ElevenLabs, Stripe) behind the first publicly documented AI-agent cold-start sale.

Andreessen Horowitz (a16z) Mon. 2026 – Mon. 2026
New Media *Menlo Park, CA*
- Built application infrastructure for a16z fellowship programs end to end (intake, review, and tooling).

Monitoring the Situation (MTS) Mon. 2026 – Mon. 2026
Host *San Francisco, CA*
- Hosted a daily TV show on X covering tech and current events.

Khosla Ventures Jan. 2026 – Mar. 2026
Technical Intern to Vinod Khosla *Menlo Park, CA*
- Built internal platform tooling and AI/ML pipelines for portfolio evaluation; worked directly with Vinod Khosla on infrastructure and investment. TypeScript, Drizzle ORM, PostgreSQL, AWS (Lambda, S3, SQS).

Human Behavior (YC X25) Jun. 2025 – Sep. 2025
Founding Engineer *San Francisco, CA*
- First hire; took the core product and platform from 0 to 1 across frontend, backend, AI/ML, design, and product. TypeScript, Postgres, LangChain, React Native, Node.js.

Jane Street Mar. 2025
First-Year Trading & Technology Program *New York, NY*

Harvard Visual Attention Lab Jun. 2023 – Aug. 2023
Research Intern *Cambridge, MA*
- Developed algorithms extracting predictive signal from unskilled crowd data for breast cancer detection; adopted by a medical data-labelling company, improving prediction accuracy 21%.

University of Virginia Biocomplexity Institute Aug. 2021 – Jul. 2023
Algorithms Developer Intern *Remote*
- Designed hybrid parallel merge-sort algorithms (Python, OpenMPI, multiprocessing) achieving 34x speedups; co-authored preprint *Approaches to the Parallelization of Merge Sort in Python* with Prof. Gregor von Laszewski.

Projects

LeMMA: Test-Time Meta-Learning for Theorem-Proving LLMs | *PyTorch, vLLM, Lean 4*
- Bilevel-optimization RL algorithm (GRPO) that trains an LLM to generate and weight problem-specific synthetic lemmas at inference time, then adapts on them before proving. Beat DeepSeek-Prover-V2 by 43% on miniF2F.
- LoRA + mixed-mode bilevel differentiation for memory-efficient meta-gradients on GPU.

BugBasher: An AI Agent Running a Business | *Devin, Ramp, Twilio, ElevenLabs, Stripe*
- Agent made 10,929 cold calls, self-revised its calling prompt from 252 to 2,793 words, escalated blockers by finding our emails in git history, and closed its first sale. Co-wrote the write-up *Make money, make no mistakes.*

Stanford Programming Contest (ProCo) | *Co-Director, Problem-Writing Lead* | *Python, C++, Polygon*
- Wrote 15+ algorithmic problems (DP, graphs, combinatorics, math) for 300+ Bay Area students; built generators and validators and deployed via Polygon. Problem sets: 2025, 2026.

XiaoPrompt (Winner, HackMIT) | *Next.js, Supabase*
- Prompt-distillation API cutting token usage 53% at 94% output consistency; A/B tested via an image-contains-text framework.

Awards & Skills

Awards: USACO Platinum; 4x AIME Qualifier; 2x CyberPatriot National Finalist & Boeing Champion; NCSF National Cyber Scholar; PROMYS 2022; Canada/USA Mathcamp 2021

Languages: Python, C++, C, TypeScript, Java, SQL, Bash, Lean 4

ML / Infra: PyTorch, vLLM, LangGraph, scikit-learn, NumPy, pandas, React, Next.js, Node.js, PostgreSQL, Drizzle, Supabase, AWS, Vercel, OpenMPI